when you are uncertain what will happen. “Making that decision,” wrote Bernstein, “is the essential first step in any effort to manage risk.”[9]

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Although Pascal and Fermat were both credited with developing probability theory, another mathematician, Thomas Bayes, wrote the piece that laid the groundwork for putting probability theory to practical action.

Born in England in 1701, exactly 100 years after Fermat and 78 years after Pascal, Bayes lived an unremarkable life. He was a member of the Royal Society, but published nothing in mathematics during his lifetime. After his death, his paper titled “Essays Towards Solving a Problem in the Doctrine of Chances” appeared. At the time, no one thought much of it. However, according to Bernstein, the essay was a “strikingly original piece of work that immortalized Bayes among statisticians, economists, and other social scientists.”[10] It provides a way for investors to make use of the mathematical theory of probability.

Bayesian analysis gives us a logical way to consider a set of outcomes of which all are possible but only one will actually occur. It is conceptually a simple procedure. We begin by assigning a probability to each of the outcomes, on the basis of whatever evidence is then available. If additional evidence becomes available, the initial probability is revised to reflect the new information. Bayes’s theorem gives a mathematical procedure for updating our original beliefs, thus changing our relevant odds.

How does this work? Let’s imagine that you and a friend have spent the afternoon playing your favorite board game, and now, at the end of the game, you are chatting about this or that. Something your friend says leads you to make a friendly wager: that with one roll of a die from the game, you will get a 6. Straight odds are one in six, a 16 percent probability. But then suppose your friend rolls the die, quickly covers it with his hand, and takes a peek. “I can tell you this much,” he says; “it’s an even number.” Now you have new information and your odds change dramatically to one in three, a 33 percent probability. While you are considering whether to change your bet, your friend teasingly adds: “And it is not a 4.” With this additional bit of information, your odds have changed again, to one in two, a 50 percent probability.